CS3244 Project Proposal
Group 6

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To help you plan for your project for timely completion, please fill out the following details on what you plan to do as a group. We have included some prompts for you to consider in your response to the various sections.

Dataset chosen and description Write in your own words (a few sentences) what this dataset is about.

1. *Does your description specify features and instances?*
2. *Do you describe the source and reliability of the data?*

The dataset used in this project contains electricity consumption and billing records from the Tunisian Company of Electricity and Gas (STEG), a public utility responsible for distributing electricity and gas across Tunisia. The company has reported significant financial losses (approximately 200 million Tunisian Dinars) due to fraudulent manipulation of electricity meters by consumers.

The dataset consists of multiple files containing information about clients and their electricity invoices. The client dataset contains one instance per client account and includes a target label indicating whether the client has been identified as fraudulent or non-fraudulent. In addition, the invoice dataset contains detailed billing records for each client over time. Since each client may have multiple invoices, the dataset contains a one-to-many relationship between clients and invoices.

Project Title Please be concise, relevant and descriptive.

Machine Learning Approaches for Electricity Fraud Detection in Imbalanced Data

Motivation Explain why this project is interesting and important and the problem you aim to address.

1. *Does your motivation clearly describe a problem?*
2. *Does it justify the problem's significance? What are the benefits of addressing this problem? Who benefits from solving it?*

Electricity fraud is a significant issue faced by utility companies worldwide. Consumers may tamper with electricity meters or manipulate usage records in order to avoid paying the correct charges. This leads to large financial losses for providers and may ultimately increase costs for legitimate consumers.

Detecting fraudulent behaviour is challenging because fraudulent cases typically represent only a small fraction of all customers, resulting in highly imbalanced datasets. Traditional rule-based systems often struggle to capture complex patterns in electricity consumption behaviour and may fail to detect subtle forms of fraud.

Machine learning techniques offer the ability to analyse large volumes of billing data and identify abnormal consumption patterns that may indicate fraudulent behaviour. By learning patterns from historical billing records, machine learning models can potentially improve fraud detection performance compared to manual inspection or rule-based approaches.

In this project, we aim to evaluate how different machine learning models perform in detecting electricity fraud and investigate how different modelling approaches and imbalance-handling strategies affect detection performance.

General Approach A high-level description of the general approach you'll use to address the questions. Sketch out how you plan to run the necessary analysis and experiments.

1. *What steps will you take in tackling your project?*
2. *For instance, how will you perform data analysis/cleaning? What machine learning models will you consider, and why do you think they are suitable for the problem?*

First, we will perform data preprocessing and aggregation to prepare the dataset for machine learning. The raw dataset contains invoice-level billing records for each client. These records will be aggregated to generate client-level features that capture electricity consumption behaviour over time. Feature engineering will include statistical summaries that may indicate abnormal patterns.

After preprocessing, the dataset will be split into training and testing sets to allow unbiased evaluation of model performance. The training data will be used for model development and hyperparameter tuning, while the test set will be reserved for final evaluation.

To ensure reproducibility and consistent experimentation, we will implement an experiment pipeline that allows different models and hyperparameter configurations to be evaluated in a systematic way. Hyperparameter tuning will be performed using cross-validation with GridSearchCV.

We will train and compare several machine learning models, including:

- Logistic Regression (baseline linear model)
- K-Nearest Neighbours (distance-based model)
- Support Vector Machines
- Decision Tree-based models

These models were chosen because they represent different classes of machine learning algorithms and allow us to compare how linear, distance-based, and non-linear methods perform on fraud detection tasks.

Because fraud detection datasets are typically highly imbalanced, we will also investigate techniques to address class imbalance, such as class weighting and other imbalance-handling strategies.

Through systematic experimentation, we will analyse how model choice, feature engineering, and imbalance handling affect fraud detection performance.

Evaluation Indicate how you will evaluate your project, i.e., how will you evaluate how your experiments have turned out?

1. *How will you evaluate how well you did as a whole? Of course the most important aspect is evaluating how well the models you proposed perform. How will you do so? What evaluation metrics will you consider?*

Since the dataset is highly imbalanced, overall accuracy is not an appropriate metric for evaluation. Instead, we will use evaluation metrics that better capture performance on imbalanced classification problems.

During model development and hyperparameter tuning, models will be evaluated using Precision-Recall Area Under the Curve (PR-AUC), implemented as the Average Precision score. This metric evaluates how well models rank fraudulent clients relative to non-fraudulent clients without requiring a fixed classification threshold.

After training the best models, we will select a decision threshold to convert predicted probabilities into fraud predictions. Threshold selection will be performed using the training data to optimise metrics such as the F1-score, which balances precision and recall.

Final evaluation will be conducted on a held-out test set using metrics including:

- Precision
- Recall
- F1-score
- Precision-Recall AUC
- Confusion matrix analysis

Particular emphasis will be placed on recall for the fraud class, since failing to detect fraudulent clients may be more costly than incorrectly flagging normal clients. We will also analyse Precision-Recall curves to better understand the trade-off between precision and recall across different models and threshold choices.

Resources A list of resources you need to conduct the project. This includes additional reading, software, compute, additional datasets, reference code (GitHub links etc) beyond your chosen dataset.

1. *If the resources are public, provide references. If they are not public, how will you plan to obtain them? For example, will the team plan to scrape the data off the internet themselves?*

Dataset:

- STEG Electricity Fraud Detection Dataset (Kaggle)

Software and tools:

- Python
- scikit-learn for machine learning models'
- pandas and NumPy for data processing
- matplotlib and seaborn for visualisation

Role Assignment and Schedule. Provide a schedule of work for the entire team indicating when you plan to complete components of the project. The schedule should have two parts with the first part to be completed by the progress report deadline of 29 March 2026 and the second part by the final report deadline of 19 April. Make sure the schedule is plausible.

1. *A tentative schedule indicating when the team plans to complete major milestones.*
2. *This must include assignment of roles to different members. For example, Alan is in charge of data cleaning, Sarah is in charge of initial model exploration. It is fine for multiple members to work on the same tasks, as long as there is a fair distribution of work. The purpose of this is to avoid cases of freeloading.*
3. *Indicate the dates of the milestones you expect to be completed by then in the next section.*
4. *Make sure you have enough time in your team schedule at the end to gather everyone's work together to gather insights through discussion and by comparing and contrasting the datasets and algorithms.*

Each member is responsible for a primary task while collaborating to ensure models are trained on the same dataset and evaluated using a common pipeline.

1. **Data Preprocessing & Feature Engineering:** Responsible for dataset exploration, data cleaning, and feature engineering. This includes identifying useful variables that may help discriminate between fraudulent and non-fraudulent transactions and ensuring the dataset is consistently prepared for all experiments.
2. **Imbalanced Data Strategy:** Responsible for researching and implementing techniques to handle class imbalance.
3. **Evaluation & Experiment Pipeline:** Responsible for implementing the experiment pipeline and choosing an appropriate evaluation framework.
4. **Tree-Based Models:** Responsible for implementing and experimenting with tree-based models, beginning with Decision Trees.
5. **Linear Models:** Responsible for implementing and experimenting with linear models including Logistic Regression and Support Vector Machines.
6. **Distance-Based Models Lead:** Responsible for implementing and experimenting with the K-Nearest Neighbours model.

All members will collaborate during evaluation, comparison of models, and report preparation.

Team Milestones

Milestone	Deadline
Phase 1: Planning & Baseline Development Dataset exploration and feature engineering Research on imbalance handling techniques Implementation of the evaluation and experiment pipeline Implementation of baseline models Preliminary evaluation of model performance Internal discussion and preparation of the progress report	29 March (before progress report)
Phase 2: Model Improvement & Final Analysis Refinement of feature engineering based on preliminary results Implementation of imbalance handling techniques Hyperparameter tuning and model improvement Evaluation and comparison of model performance Discussion of findings and consolidation of results Preparation of final report and analysis	19 April (before final report)

Individual Milestones

Name	Milestone	Deadline
Thang	Completion of dataset exploration and initial feature engineering	15 March (Sun)
Phuong	Literature review on imbalanced data handling techniques	15 March (Sun)
Bryan	Implement an appropriate experiment & evaluation pipeline	15 March (Sun)
Mavis	Implementation and baseline experimentation for KNN	22 March (Sun)
Alyssa	Implementation and baseline experimentation for linear models (Logistic Regression and SVM)	22 March (Sun)
Alfreda	Implementation and baseline experimentation for Tree Models (Decision Tree)	22 March (Sun)
Everyone	Team meeting to review preliminary results, discuss findings, and prepare the progress report	29 March (Sun)

Individual milestones are specified only for Phase 1. Milestones for Phase 2 will be planned after the progress report once the team has evaluated the results of the baseline experiments.